

SNAPFRAME · REQUIREMENTS

SnapFrame requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	R5, R7, R8 and R11 restated and R6 and R9 status notes updated after Amish's decisions of 2026-10-02 (SNF-DEC-001)
0.6	2026-10-01	Amish Chadha	Status from SNF-CAL-001 v0.4 after the design for construction (SNF-DDR-003); R10 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply SNF-DDR-001 (R10 redefined as the frame kit; 1 in rafters); status from SNF-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

SnapFrame requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after co-design sessions. The status column comes from the TRL 3 calculation note SNF-CAL-001 v0.4, for the constructable size M frame of SNF-DDR-003 with 1 in EMT rafters (SNF-DDR-001 D2) and ridge tubes (SNF-DDR-002 D8), 3/4 in EMT posts and eave tubes, and one eave node variant (SNF-DDR-002 D9). Two requirements are **not met**: R6 (wind) and R13 (fire). Two are at risk: R8 (anchors) and R9 (nodes). R1, R2, R3, R5, R7, R11 and R12 are met, R5, R7 and R11 as restated by Amish on 2026-10-02 (SNF-DEC-001): three packages, a third tarpaulin for the front gable, and the carried mass counted on the frame kit alone. R8 was restated the same day (1.5 kN at the two rear corner feet). R10 is reported against its value-engineering target: the constructable frame kit is \$24.14 over it.

ID	Requirement	Target	Status (SNF-CAL-001)	Verification (TRL 3 or later)
R1	Give enough covered living space	Size M: 16 m ² floor or more, which is 3.5 m ² per person for four people (Sphere minimum)	Met: 4.0 x 4.0 m = 16.0 m ²	Geometry
R2	Give standing headroom	2.0 m or more over at least 60 % of the floor	Met: 73 % of the floor, measured to the rafter underside	Geometry
R3	Cover several sizes from one design	At least three sizes (S about 9 m ² , M 16 m ² , L about 24 m ²) generated from one parametric node model; M and L share every tube length	Met: <code>cad/src/model.py</code> generates and exports the S, M and L node sets	Parametric model
R4	Go up without tools	Two adults with no tools and no prior training, using a picture guide, erect frame and skin of size M in 60 min or less	Not verifiable at TRL 3: estimate about 46 min	Time study in a later user trial (TRL 4 or later)
R5	Ship flat	No member longer than 2.1 m; all members straight; kit in three packages (two tube bundles and a bag), each 25 kg (55 lb) or less and 0.10 m ³ or less (restated 2026-10-02, SNF-DEC-001)	Met on paper: longest member 2.064 m; tube bundles about 16.5 and 12.1 kg; bag 14.7 kg without tarpaulins, which are issued from agency stock	Massing model, then weighing
R6	Resist wind with the skin on	Every member stays elastic with a safety factor of 1.5 or more on yield in gusts of 20 m/s (72 km/h, 45 mph)	Not met: rating about 19.7 m/s (71 km/h, 44 mph) with 1 in ridge tubes (SNF-DDR-002 D8); 3/4 in posts reach 1.46 at 20 m/s; 1 in rafters 1.66 and 1 in ridge tubes 2.22. The frame analysis is run now (decided 2026-10-02); if it has not been run when the prototype tubes are bought, 1 in posts are fitted	Wind load calculation (CAL), then frame analysis
R7	Fit standard relief tarpaulins	Two 4 x 6 m tarpaulins close the roof, both side walls and the rear gable, and part of a third, cut to include a door flap, closes the front gable; all from agency stock (restated 2026-10-02, SNF-DEC-001)	Met on paper by area: full enclosure needs about 51.4 m ² against 72 m ² in three tarpaulins; the cutting plan with the door flap and the corner cable hem is still to be drawn	Cutting and folding plan
R8	Anchor without a hammer	Every foot and guy anchored by hand; each anchor holds 1.0 kN uplift in firm soil, and each of the two rear corner feet 1.5 kN with longer anchors (raised 2026-10-02, SNF-DEC-001)	At risk: estimated demand up to 1.28 kN at a rear corner foot before cable pretension (the gable and side wall cables clip to that anchor's eye), inside the 1.5 kN target there; holding capacity not verifiable at TRL 3	Anchor pull-out data for the chosen anchor and soils
R9	Keep nodes strong in sun and cold	Nodes carry the socket forces from R6 with a factor of 2 after two years outdoors, from -10 to 70 °C surface temperature	At risk: hitch pin bearing on the socket about 32 MPa with the factor of 2; polymer chosen on 2026-10-02 (glass- or carbon-filled PA12-class nylon, SNF-DEC-001), printed coupons to be tested for pin bearing at 70 °C at TRL 4	Material data, then node load tests (TRL 4)

ID	Requirement	Target	Status (SNF-CAL-001)	Verification (TRL 3 or later)
R10	Keep to the value-engineering target	Size M frame kit at or below the \$445 value-engineering target (<code>budget_usd</code> , a control target, not a limit; Amish, 2026-10-01) in prototype quantities, excluding tarpaulins, which come from agency stock and are costed separately (SNF-DDR-001 D1; SNF-DDR-002)	Frame kit about \$469 for the constructable design, \$24.14 over the value-engineering target; tarpaulins about \$50 extra	Priced BOM
R11	Be carried by two people	Complete size M frame kit, 50 kg (110 lb) or less; tarpaulins are issued separately from agency stock and not counted (restated 2026-10-02, SNF-DEC-001)	Met on paper: frame kit about 43.3 kg (52.5 kg with two tarpaulins)	Massing model, then weighing
R12	Be repairable in the field	Any member or node replaced by hand in 5 min or less; four tube types and three node types only	Met: four tube types; three node types in four printed variants, none handed (one eave node with a blank capped socket at the corners, SNF-DDR-002 D9). Replacement time not verifiable at TRL 3	Design review
R13	Limit fire spread	Skin meets a recognized flame spread test for tents (for example CPAI-84) or the kit ships with a fire-retardant tarpaulin option	Not met with standard polyethylene relief tarpaulins, which are usually not flame retardant	Material data for the chosen tarpaulin

■ Assumptions

- Reference size M (SNF-DDR-001 D6): 4.0 x 4.0 m floor, two 2.0 m bays, eave 1.8 m, ridge 2.6 m (roof pitch 21.8°), three gable frames.
- Tube (SNF-DDR-001 D2, SNF-DDR-002 D8): 1 in EMT rafters and ridge tubes (OD 29.54 mm, wall 1.448 mm, 1.003 kg/m) and 3/4 in EMT posts and eave tubes (OD 23.42 mm, wall 1.245 mm, 0.681 kg/m), ANSI C80.3 nominal dimensions. Yield strength assumed 275 MPa, to be confirmed from supplier data.
- Wind: $q = \frac{1}{2}\rho v^2$ with $\rho = 1.225 \text{ kg/m}^3$; windward wall +0.8, leeward wall -0.5, roof suction -0.7 (first-order values in the style of ASCE/SEI 7, not a code check). Each tarpaulin panel shares its load between its four edge members by 45° tributary lines. See SNF-CAL-001 for the full list.
- Members are treated as simply supported between node centers; node fixity is ignored, which is conservative for bending.
- Snow and cyclone rating are out of scope for the first release (SNF-DDR-001 D7).
- Value-engineering target: \$445 is the `budget_usd` in `project.yaml` (raised from \$400 when Amish approved the budget on 2026-09-26), applied to one size M frame kit including anchors, cables, cable bolt sets and bags, and excluding tarpaulins (SNF-DDR-001 D1). It is a hypothetical control target, not a limit (Amish, 2026-10-01).